

Document reference		NPS / 002 / 011					
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NPS/002/011 – Technical Specification for Cable Cleats and Saddles

1. Purpose

The purpose of this document is to detail the requirements of CE Electric UK for cable cleats and saddles used to support and fix various sizes and types of cables utilised on the companies' distribution system.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NPS/002/011	1 (June 06)	Specification for Cable Cleats and Saddles

2. Scope

This specification details the requirements for Trefoil cable cleats for voltage range 11kV to 20kV and single/double fixing cleats and saddles for single and 3-core cables of voltage range 230/400volts.

3. Requirements

3.1 Trefoil Cable Cleat

Trefoil cable cleats shall be so designed to fix, retain and support cables to wood poles, steel structures etc. and shall be capable of supporting the cables during a fault / short circuit conditions.

Metal cleats shall be range taking, hot dipped galvanised in accordance with BS 7371- 6 and shall be either single or two bolt fixing, with the plastic fillers/pads being manufactured from an LSF material.

The cable sizes to be covered are detailed in attached Appendix.

3.2 Single Fixing Cleat

Single fixing cleat, shall be manufactured from a flame retardant polypropylene (or equivalent material), and so designed to secure and support the cables and resist the mechanical and electrical loads of the distribution system.

After installation the cleats can be expected to be subjected to the full range of climatic conditions encountered in the UK, and may be exposed to sunlight for significant periods. Cleats may be installed onto wood poles and therefore in contact with the pole preservation medium such as creosote.

The cable sizes to be covered are detailed in attached Appendix.

3.3 Single way Snap on Saddle

Single way two bolt fixing snap on saddle clips, for securing power cables and uPVC service duct. The shoulders of the saddle shall be so designed to hold the cable or ducting away from the mounting surface.

After installation the cleats can be expected to be subjected to the full range of climatic conditions encountered in the UK, and may be exposed to sunlight for a significant period. Cleats may be installed onto wood poles and therefore in contact with the pole preservation medium such as creosote.

The cable sizes to be covered are detailed in attached Appendix.

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3.4 Pole Cleats

Heavy duty pole cleats designed to secure power cables onto wood poles. The cleats mounting surface shall be curved and suitable for fixing to a typical 200mm diameter pole. Cleats installed onto wood poles will be in contact with the pole preservation medium such as creosote.

After installation the cleats can be expected to be subjected to the full range of climatic conditions encountered in the UK, and may be exposed to sunlight for a significant period.

3.5 Single Fixing Cable Cleat and Spacer

Range-taking two way stacking cleat suitable for securing power cables onto wood poles and flat surfaces. A requirement also exists for a suitable spacer, which would allow multiple cables to be stacked. The cleat and spacer shall be suitable for an M10 coach screw type fixing.

After installation the cleats can be expected to be subjected to the full range of climatic conditions encountered in the UK, and may be exposed to sunlight for a significant period. Cleats may be installed onto wood poles and therefore in contact with the pole preservation medium such as creosote.

3.6 One and Two hole Cable Clamp

The cable clamps shall be manufactured from black flame retardant zero halogen phosphorus free nylon (LSF) or an equivalent material, and so designed to secure and support the cables onto flat surfaces and resist the mechanical and electrical loads of the Distribution system.

The cable sizes to be covered are detailed in attached Appendix.

3.7 Washers

Manufactured from black polypropylene or a similar material, two types of washers are required, one to act as a "cushion" and one to "stand-off" signs and notices, which are to be permanently attached to wooded poles, sub-station doors etc.

Requirements are as detailed in the attached Appendix.

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4. References

4.1 External Documentation

Reference	Title
BS 7371-6 1998	Coatings on metal fasteners. Specification for hot dipped galvanized coatings

4.2 Internal documentation

Reference	Title
Nil	

5. Definitions

Term	Definition
Nil	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
V. Gorny	Standards Engineer	V. Gorny	22/06/09

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G. Bolden	Procurement Manager	G. Bolden	29/06/09

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
R. McMahon	Standards Engineer	R. McMahon	23/06/09

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S. Johnson	DBD Administrator	S. Johnson	07/07/09

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C. Holdsworth	Standards and Assessments Manager	C. Holdsworth	29/06/09

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	29/06/09

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Appendix 1 – Cable Range/Types

Section 3.1	Cable Type	Approximate Cable Diameter Triplex Formation	Approximate Diameter of Individual Core
	11kV – 95mm Triplex Solid Aluminium Core	60.5mm	28.1mm
	11kV – 185mm Triplex Solid Aluminium Core	68.1mm	31.6mm
	11kV – 300mm Triplex Solid Aluminium Core	80.4mm	37.3mm
	11kV – 300mm Triplex Stranded Copper Core	80.6mm	37.4mm
	20kV – 95mm Triplex Solid Aluminium Core	68.4mm	31.7mm
	20kV – 185mm Triplex Solid Aluminium Core	78.15mm	36.27mm

Section 3.2	Cable Type	Approximate Cable Diameter
	16mm CNE Single Phase Stranded Copper Core	12.5mm
	16mm SNE Single Phase Stranded Copper Core	15mm
	25mm CNE Single Phase Solid Aluminium Core	13mm
	25mm SNE Single Phase Solid Aluminium Core	14.8mm
	35mm CNE Single Phase Solid Aluminium Core	14.4mm
	35mm SNE Single Phase Solid Aluminium Core	18.1mm
	25mm CNE 3-Core Solid Aluminium	22.6mm
	35mm CNE 3-Core Solid Aluminium	24.8mm
	35mm CNE LSF Single Phase Solid Aluminium Core	13.5mm

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Section 3.3 & 3.4	Cable Type	Approximate Cable Diameter
	95mm Waveform: 3-Core Solid Aluminium	34.2mm
	95mm Waveform: 4-Core Solid Aluminium	38.3mm
	185mm Waveform: 3-Core Solid Aluminium	46.6mm
	185mm Waveform: 4-Core Solid Aluminium	53.2mm
	185mm LSF Waveform: 3-Core Solid Aluminium	46.6mm
	300mm Waveform: 3-Core Solid Aluminium	56mm
	300mm Waveform: 4-Core Solid Aluminium	63mm

Section 3.5	Cable Type	Approximate Cable Diameter
	35mm Double Insulated Single Core Copper Conductor	12.1mm
	95mm Double Insulated Single Core Copper Conductor	18.3mm

Section 3.6	Cable Type	Approximate Cable Diameter
	One Hole Cable Clamp For Cable Range	19mm – 23mm with 10mm fixing hole
	Two Hole Cable Clamp For Cable Range:	57mm – 64mm with 10mm fixing holes
		70mm – 76mm with 10mm fixing holes

Section 3.7	Requirements
	Washer: 14mm x 2mm thick x 5mm fixing hole
	Washer: 14mm x 6mm thick x 5mm fixing hole

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Appendix 2 – Addendum to Supplier Requirements

To enable CE Electric UK to install the product(s) in accordance with the manufacturer's recommendations, if appropriate, the supplier shall provide a single copy of a drawing, descriptive leaflet and instruction manuals appropriate to the goods being offered, which shall incorporate details of the recommended installation procedures required in order to provide optimal performance of the goods during their operational life.

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Appendix 3 – Logistical Requirements

The supplier shall ensure that each item is suitably packaged and protected to maintain the product and packaging as “fit for service” prior to installation taking account of the potential for an outdoor storage environment.

All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Where product packages tendered are made up of sub-packages all the sub-packages shall unless varied by this specification, be supplied securely packaged together. Where items are provided in bagged/boxed form the material from which the bags are manufactured shall be capable of sustaining the package weight and resisting puncture by the materials within. Suppliers shall submit at the time of tendering the details of the proposed packing (i.e. materials composition and structure) to be used for each product. Where the supplier is unable to provide packaging suitable for outdoor storage then this should be stated at the time of tender.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple package of like products. Where products packages tendered are made up of sub packages each sub package shall be marked. As a minimum requirement the following shall be included: -

- Manufacturer's trademark or name
- Supplier's trademark or name
- Description of item
- Date of packaging and/or batch number
- CE Electric UK brand name
- CE Electric UK product code
- Package weight